

Unit 5 Pre-Test

The Pythagorean Theorem

Standards assessed: NC.8.G.6, NC.8.G.7, NC.8.G.8

Name: _____ Date: _____ Period: _____

SCORE **3.0** Basic Understanding

NC.8.G.7 Apply the Pythagorean theorem to find the hypotenuse.

• I can use $a^2 + b^2 = c^2$ to find the hypotenuse of a right triangle.

Match each vocabulary term to its definition, then find each hypotenuse.

– ▼ hypotenuse	A. The two sides that form the right angle.
– ▼ legs	B. $a^2 + b^2 = c^2$, relating the sides of a right triangle.
– ▼ Pythagorean theorem	C. The side opposite the right angle; the longest side.
– ▼ right triangle	D. A triangle that contains one 90° angle.
– ▼ square root	E. A number that, multiplied by itself, gives a given value.
– ▼ converse	F. If $a^2 + b^2 = c^2$, then the triangle is a right triangle.

Pythagorean theorem — reference

$$a^2 + b^2 = c^2$$

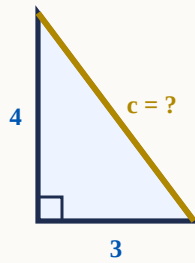
c is the hypotenuse (opposite the right angle); *a* and *b* are the legs.

1. Find *c*

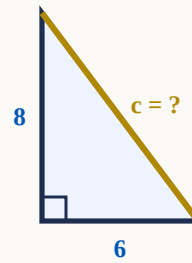
Legs 3 and 4. Find the hypotenuse.

2. Find *c*

Legs 6 and 8. Find the hypotenuse.



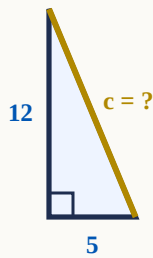
$c =$ _____



$c =$ _____

3. Find c

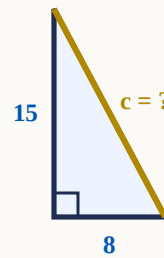
Legs 5 and 12. Find the hypotenuse.



$c =$ _____

4. Find c

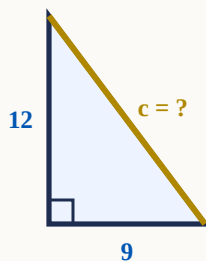
Legs 8 and 15. Find the hypotenuse.



$c =$ _____

5. Find c

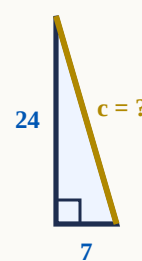
Legs 9 and 12. Find the hypotenuse.



$c =$ _____

6. Find c

Legs 7 and 24. Find the hypotenuse.



$c =$ _____

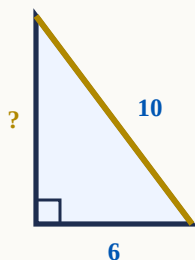
NC.8.G.7 Find a missing leg and estimate non-perfect square roots.

• *I can find a missing leg and round answers to the nearest tenth when needed.*

Find each missing side. Round to the nearest tenth where asked.

7. Find a leg

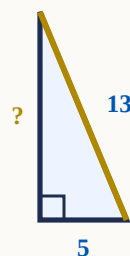
One leg 6, hypotenuse 10. Find the other leg.



a =

8. Find a leg

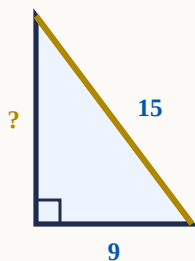
One leg 5, hypotenuse 13. Find the other leg.



a =

9. Find a leg

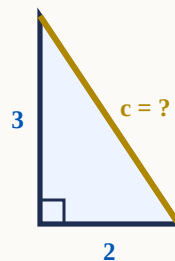
One leg 9, hypotenuse 15. Find the other leg.



a =

10. Nearest tenth

Legs 2 and 3. Find the hypotenuse to the nearest tenth.



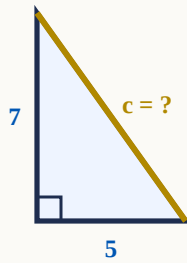
c ≈

11. Nearest tenth

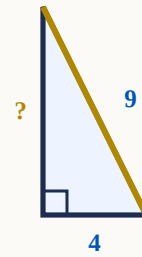
Legs 5 and 7. Find the hypotenuse to the nearest tenth.

12. Nearest tenth

One leg 4, hypotenuse 9. Find the other leg to the nearest tenth.



$c \approx$



$a \approx$

SCORE 5.0 Advanced Content

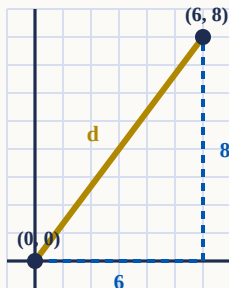
NC.8.G.8 Apply the theorem to the coordinate plane and real-world problems.

- I can find the distance between two points and solve real-world right-triangle problems.*

Solve each problem.

13. Distance

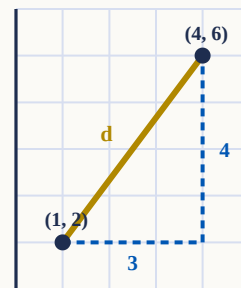
Find the distance between $(0, 0)$ and $(6, 8)$.



$d =$

14. Distance

Find the distance between $(1, 2)$ and $(4, 6)$.



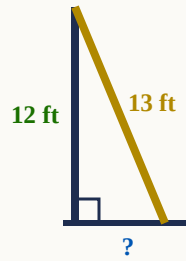
$d =$

15. Apply

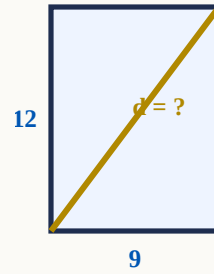
A 13 ft ladder reaches 12 ft up a wall.
How far is the base from the wall?

16. Apply

A rectangle is 9 by 12. Find the length
of its diagonal.



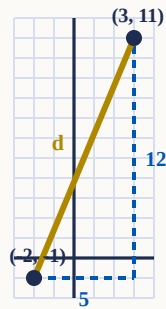
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d =

17. Distance

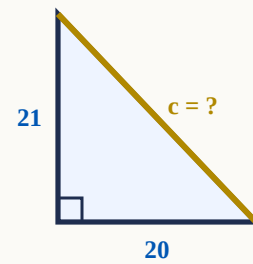
Find the distance between $(-2, -1)$ and $(3, 11)$.



d =

18. Apply

A park is a right triangle with legs 20 m and 21 m. Find the longest side.



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